Justifying constructions using the pro[illegible]ies of quadrilaterals

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a

To prove: $\overleftrightarrow{AB}$ **is constructed parallel to** $\overleftrightarrow{DC}$

	Statements	Reasons
1.	$\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$	The arcs that created the segments have the same radius.
2.	$ABCD$ is a paralellogram	If a quadrilateral has both pairs of opposite sides congruent, then it is a parallelogram.
3.	$\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$	Definition of a parallelogram.

b

To prove: $\overleftrightarrow{CD}$ **is constructed so that** $\overline{AB} \perp \overline{CD}$

	Statements	Reasons
1.	$\overline{AC} \cong \overline{CB} \cong \overline{BD} \cong \overline{DA}$	The arcs that created the segments have the same radius.
2.	$\overline{ACBD}$ is a rhombus	Definition of a rhombus
3.	$\overline{AB} \perp \overline{CD}$	If a quadrilateral is a rhombus, then its diagonals are perpendicular.

c

To prove: $\overleftrightarrow{CD}$ **is constructed so that** $\overline{AE} \cong \overline{BE}$

	Statements	Reasons
1.	$\overline{AC} \cong \overline{CB} \cong \overline{BD} \cong \overline{DA}$	The arcs that created the segments have the same radius.
2.	$ACBD$ is a paralellogram	If a quadrilateral has both pairs of opposite sides congruent, then it is a parallelogram.
3.	$\overline{CD}$ bisects $\overline{AB}$	If a quadrilateral is a parallelogram, then its diagonals bisect one another.
4.	$\overline{AE} \cong \overline{BE}$	Definition of a bisector

d

To prove: $\overline{PR}$ is const[illegible]d so that $\angle TPR \cong \angle RPQ$

	Statements	Reasons
1.	$\overline{PT} \cong \overline{TR} \cong \overline{RS} \cong \overline{SP}$	The arcs that created the segments have the same radius.
2.	$PTRS$ is a rhombus	Definition of a rhombus
3.	$\overline{RP}$ is a line bisector.	If a parallelogram is a rhombus, then each diagonal bisects a pair of opposite angles.
4.	$\angle TPR \cong \angle RPQ$	Definition of an angle bisector

e

To prove: $\overline{PR}$ is constructed from line segment $\overline{PQ}$ such that $m\angle RPS = 30^\circ$

	Statements	Reasons
1.	$\overline{PT} \cong \overline{TS} \cong \overline{SP}$	The arcs that created the segments have the same radius.
2.	$\triangle PTS$ is a equilateral	Definition of an equilateral triangle
3.	$m\angle TPS = 60^\circ$	Each angle of an equilateral triangle measures 60°
4.	$\overline{PT} \cong \overline{TR} \cong \overline{RS} \cong PS$	The arcs that created the segments have the same radius.
5.	$PTRS$ is a rhombus	Definition of a rhombus
6.	$\overline{RP}$ is an angle bisector	If a parallelogram is a rhombus, then each diagonal bisects a pair of opposite angles.
7.	$\angle TPR \cong \angle RPS$	Definition of an angle bisector
8.	$m\angle TPR = m\angle RPS$	Definition of congruence
9.	$m\angle TPR + m\angle RPS = m\angle TPS$	Angle addition postulate
10.	$m\angle RPS + m\angle RPS = m\angle TPS$	Substitution
11.	$m\angle RPS + m\angle RPS = 60^\circ$	Substitution
12.	$2m\angle RPS = 60^\circ$	Simplify
13.	$m\angle RPS = 30^\circ$	Division property of equality

2 A point A is arbitrarily chosen along the circle. We create the diameter $\overline{AC}$ by passing it through the center of the circle at O. We choose a sufficiently large radius and construct the pair of arcs at E and F by centering them at A and C. Then the line segments $\overline{AE}$, $\overline{CE}$, $\overline{CD}$ and $\overline{DA}$ are congruent since they have been constructed from the same radius. So $AECF$ is a rhombus.

We then construct $\overline{EF}$. The diagonals of a rhombus are perpendicular, so $\overline{EF}$ is perpendicular to $\overline{AC}$. $AECF$ is a rhombus, which means it's also a parallelogram. The diagonals of a parallelogram bisect one another, so $\overline{EF}$ bisects $\overline{AC}$. Since $\overline{AC}$ is a diameter of the circle, then EF bisects $\overline{AC}$ through the center at O.

We then extend $\overline{EF}$ to the edges of the circle at B and D. Then $\overline{BD}$ bisects and is perpendicular to $\overline{AC}$. $\overline{AC}$ also bisects $\overline{BD}$ since $\overline{BD}$ is a diameter as well. $\overline{BD}$ and $\overline{AC}$ are congruent since they are diameters of the same circle. So $ABCD$ is a square, since the diagonals are congruent, perpendicular and bisect one another. And finally, since the vertices lie on the circle, $ABCD$ is an inscribed square.